

Your Photo Gallery, but Smarter

*A Local-First Semantic Image Search System
That Runs on Your Laptop*



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*I have 40,000 photos.
I can't find any of them.*

CTRL+F doesn't work on JPEGs.

What We'll See Today



Index

A library of your own photos, embedded into a single vector space and stored in FAISS on disk.



Search

Natural-language queries - or another photo as the query itself.



Cluster

Unsupervised groups that look like the albums you never made - and labels you never wrote.

Why Local-First?

	Cloud Apps (e.g. Google Photos)	Local-First Apps
Where data lives	 Company servers	 Your device
Privacy	 Company-controlled	 Fully user-owned
Cost model	 Ongoing subscription	 One-time / minimal
Internet dependency	 Required	 None
Setup effort	 Easy	 Simple
Customisable	 Fixed features	 Flexible by design

CLIP – The Text/Image Embedder

«A dog sitting on a chair»

CLIP

[0.21, -0.44, 0.87, -0.05, ..., 0.17]

512 vector dimension



CLIP

[0.20, -0.42, 0.85, -0.04, ..., 0.18]

512 vector dimension

SHARED LATENT SPACE

image vec

text vec

Similar things end up close.

CLIP: one shared space for images and text.

Search – FAISS Index Architecture

IndexFlatIP

How similarity is computed.

- Brute-force inner-product search
- L2-normalization → cosine similarity
- Exact Nearest Neighbor Search in milliseconds- no approximation

IndexIDMap

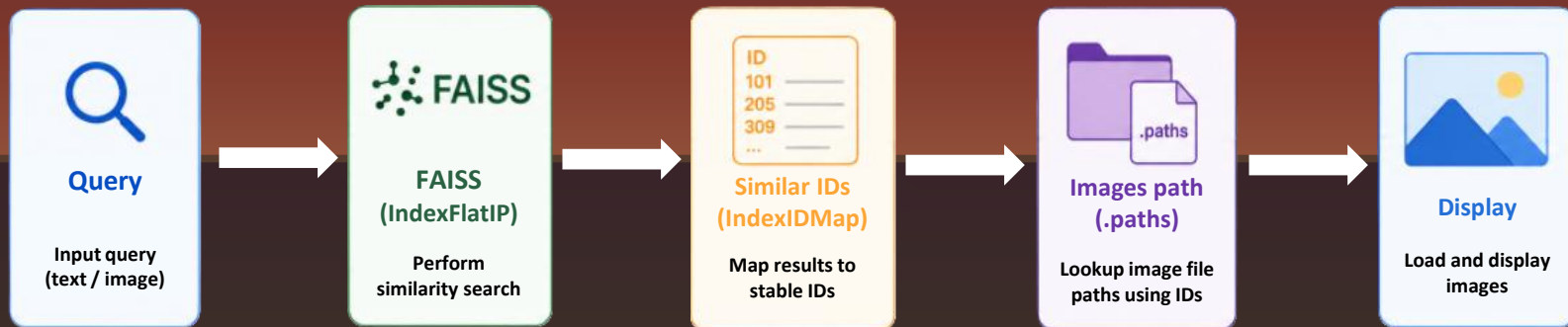
Which image a result points to.

- Wraps FAISS index to assign explicit 64-bit IDs per vector
- Add and delete vectors without shifting
- Path mapping survives mutation

.paths

The bridge to the filesystem.

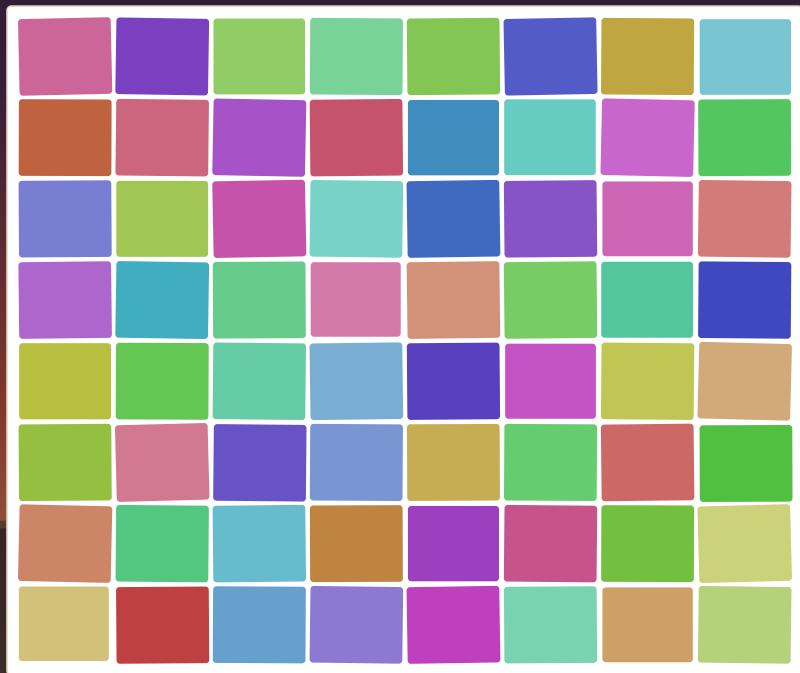
- External lookup: ID → image path
- Synced with index after addition or deletion
- Prevents duplicate indexing



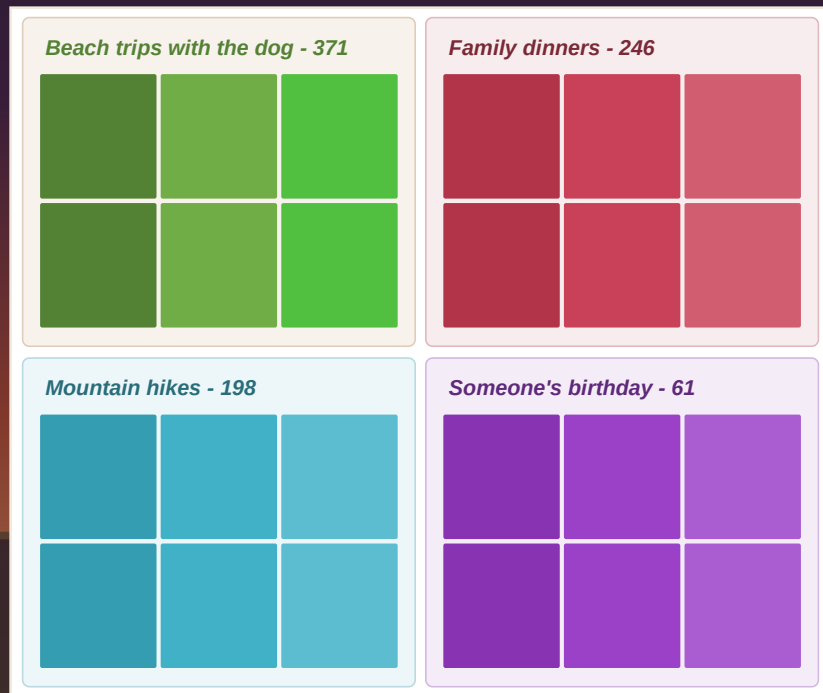
Cluster to Discover

Search finds what you ask for. Clustering shows you what you have.

Mixed chaotic photos



Well-organized photo gallery



The BERTopic Pipeline

Embed → Reduce → Cluster → Label



CLIP

One embedding
per photo, same
model as search.

CLIP produces 512-dim
vectors that capture visual
semantics.



UMAP

Preserve local
neighborhood while
lowering vector
dimensions.

UMAP compresses vectors
to a lower dimension.



HDBSCAN

Density-based
clustering.

HDBSCAN finds dense
regions → clusters.



BLIP + Ollama

Per-cluster caption
and kw → local LLM
rewrites it as a theme.

BLIP captions;
Ollama turns them into a
readable group label.



LIVE DEMO

“Pray with me” time

Key Takeaways



Local-first wins

Privacy, offline, zero cost — your data never leaves your machine



CLIP unifies modalities

One shared embedding space for images and text makes semantic search trivial



Quite simple stack, real power

FAISS + BERTopic = search + clustering



Built for your own data

No cloud API, no subscription — works with old photos, new photos, customizable as desired

What's Next



Temporal + EXIF search

Combine CLIP with image metadata (date, GPS, etc) to search for “photos from last Christmas”



File watcher auto-indexing

Watchdog monitors your photo folder and indexes new files automatically



Auto-generated photo stories

Feed cluster labels + representative photos to an LLM — get a narrative photo book



From Images to Audio

Automatically group music with same genre

Search for a particular song by mood or instruments

Thank you!

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